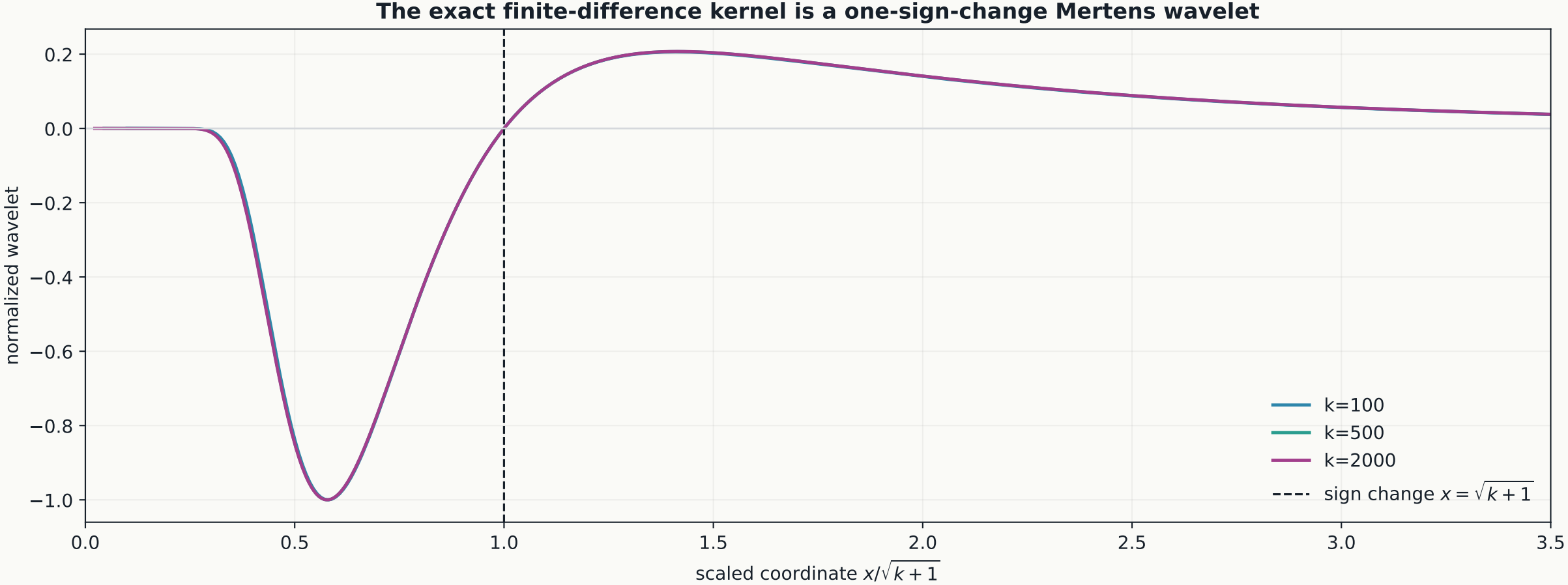
The exact finite-difference kernel is a one-sign-change Mertens wavelet
normalized wavelet
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
scaled coordinate $x/\sqrt{k+1}$
k=100
k=500
k=2000
sign change $x=\sqrt{k+1}$